AI and LLM Prompt Engineering

Student Syllabus · One-day course · 8:00–4:00

Date: [DATE] · **Room:** [ROOM / BUILDING] · **Instructor:** [NAME], [EMAIL]

About this course

The quality of results produced by generative AI often depends on the quality of the instructions it receives. Effective prompt design helps users communicate their intent clearly, provide appropriate context, establish constraints, and guide large language models (LLMs) toward more useful outputs.

In this hands-on course, participants will learn practical techniques for designing, testing, and refining prompts for workplace applications. Participants will explore how prompt structure influences AI-generated responses and how thoughtful prompt design can improve quality, consistency, and efficiency.

This offering applies those techniques to everyday staff work: writing for busy executive readers, and making recurring tasks faster. **You'll work individually** on your own laptop all day. **There is no exam and no graded work.**

Learning objectives

1. Explain how large language models use prompts and context to generate responses.
2. Write clear, structured prompts that produce more useful and relevant outputs.
3. Apply prompting techniques to improve accuracy, consistency, format, and usefulness.
4. Refine prompts through iterative testing and evaluation.
5. Use prompting strategies that improve efficiency while maintaining output quality.

The day

Time	Session	You'll practise
8:00	*Arrival and setup*	Laptop on Wi-Fi, logged in to free ChatGPT
8:30	**Welcome**	The objectives, the data rule, and naming your own recurring task
8:45	**How LLMs use prompts, and the rules of the road**	Running one prompt twice; sorting what's OK to paste
9:45	*Break*	
10:00	**The six slots**	Rebuilding a vague prompt, one slot at a time
11:00	**Writing for executives**	Turning a wordy memo into a BLUF email
12:00	*Lunch (60 min)*	
1:00	**AI in your workflow**	A chain of prompts, then version 1 of *your own* template
2:00	*Break*	
2:15	**Test and refine**	Finding errors, then testing your template against criteria
3:15	**Put it together**	A decision memo from a raw email thread
3:45	**Close**	Your 30-day plan
4:00	*End*	

What you'll leave with

A tested prompt template for one recurring task from your own job, that a colleague could use without help. You'll build it in the afternoon, test it, and refine it. You'll also take home:

Handout	What it's for
Data rules card	What may go into free ChatGPT, and into the command's approved tool at work
Six slots card	A checklist for any prompt. When the output disappoints, find the empty slot
Executive writing card	The four rules for busy readers, three prompt patterns, and a sign-off checklist
Prompt library	Eight starter templates for daily tasks: BLUF emails, action tables, status entries, decision memos, and more

The one rule for today

> **Free ChatGPT is a public tool. Only the fabricated course material goes in.** Nothing from work, even to "try it out." No student, personnel, or health records, ever. Every exercise text is invented for training, and so is the sample input you'll write for your own template.

At work afterwards, use only the command's approved AI tool, and only for what it's authorized to hold.

What to bring

- **Your own laptop** and charger. There's nothing to install
- **A free ChatGPT account** you can log in to, tested before the day. No paid features are used
- One recurring task from your job that takes too long. You'll describe it in general terms only

Certificate of completion

Attend the full day to receive a certificate of completion. Sign in at 8:30 and again at 1:00. Certificates are sent within `[N]` business days.

How the day works

- **You work on your own.** The instructor demonstrates on the projector, and in each debrief, a volunteer's prompt goes on screen.
- **Start a new chat for each test.** Earlier messages change later answers. Turn memory off, or use a temporary chat if your account has one.
- **The AI drafts, and you sign.** Check every number, date, and name.
- **Ask questions any time.** Policy questions go to `[AI / CYBERSECURITY OFFICE POC]`, and the answers come in the follow-up email.

After the course

You'll receive the handouts by email. At 30 days, we'll ask one question: did you use your template at work? Related courses in this series: *Introduction to AI and Machine Learning*, *Building an AI Agent*, *AI Ethics and Governance*, and *Data Analytics, Cleaning and Visualization in Excel*.